

SplineLoc: Map-Wide Single-Scan Localization from Rover LiDAR to Coarse Lunar DEMs

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Abstract—Long-range lunar rovers rely on dead-reckoning that drifts without bound, so periodic global fixes are needed to keep the pose estimate stable over a traverse. Orbital digital elevation models (DEMs) offer globally referenced terrain, but their coarse sampling differs substantially from dense rover LiDAR observations. We present SplineLoc, a single-scan LiDAR-to-DEM localization method that searches the full extent of a supplied DEM without position or heading initialization, terrain-landmark extraction, or a reference-scan database. Given a scan levelled using externally supplied roll and pitch, we represent the DEM and scan as cubic B-spline height fields, providing a common continuous representation for height comparison. An offset-invariant height objective decomposes into masked cross-correlations, enabling FFT evaluation over lattice translations at sampled headings. Retained candidates are then refined and re-ranked using raw LiDAR points and the analytic gradient of the map height field.

On four lunar terrains simulated from real orbital topography and six DEM resolutions from 0.25 to 10 m/pixel, one configuration achieved the highest SR@5m in all 24 site-resolution conditions. Mean SR@5m remained 91.75% at 10 m/pixel, compared with 2.75% for the strongest baseline at that resolution. Performance also remained high on the 2000×2000 m Apollo 14 and Apollo 15 maps, indicating that the method remains effective under both substantial DEM coarsening and map-wide search over large terrain regions. Replacing the simulation-derived reference maps with aligned native orbital products changed SR@5m by at most two percentage points, suggesting that localization is driven by terrain structure preserved across map representations rather than fine-scale details specific to the simulation. The code will be available on the project page: <https://anonymous1312602.github.io/SplineLoc-ICRA/>

I. INTRODUCTION

Reliable long-range lunar surface exploration requires a rover to periodically re-establish its position in a global reference frame. The Moon offers no GNSS, so dead-reckoning estimates based on wheel, visual, and inertial odometry accumulate drift over extended traverses [1]. Vision-based localization is further challenged by low-texture regolith and large illumination variations, and passive optical sensing is severely constrained in permanently shadowed regions. LiDAR, in contrast, measures terrain geometry independently of ambient illumination, and orbital digital elevation models (DEMs) provide globally referenced topography over broad regions [2], [3]. Registering rover LiDAR geometry against an orbital DEM is therefore an attractive means of periodic, illumination-independent global localization.

The practicality of this approach depends on the DEMs available. LROC Narrow Angle Camera (NAC) stereo DTMs provide roughly 2–5 m/pixel topography for selected sites [4], and specialized products reach finer scales at

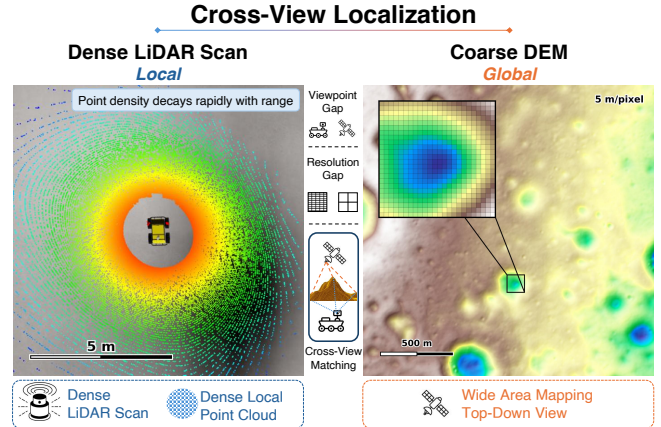


Fig. 1. **Cross-view LiDAR-to-DEM localization.** A dense rover LiDAR scan is registered to a coarse orbital DEM to recover the rover’s global planar pose. Both panels are rendered from our OmniLRS simulation environment used in the experiments.

a few locations [5]. Reference-map resolution thus varies across sites and is often far coarser than the rover’s LiDAR sampling, so a localization method for this setting should remain effective across these resolutions without site-specific redesign.

Existing global localization methods typically introduce an auxiliary structure or prior. Planetary approaches rely on extracted terrain structures such as topographic peaks, craters, or horizon profiles [6], [7], [8], accumulated observations and odometry [9], or visual appearance [10], [11], [12], [13]. General LiDAR localization searches a reference-scan database [14], [15]. Most closely related, Zhang et al. [16] match LiDAR against a DEM-derived surface within Monte Carlo localization, but initialize particles inside a predefined 20 m × 20 m region. We address the setting of localizing a single LiDAR scan over the full extent of a supplied DEM without a bounded position initialization, an externally supplied heading, terrain-landmark extraction, or a reference-scan database.

The central difficulty is the representation gap between the two sources: a LiDAR scan is irregularly sampled, range-dependent, and only partially observes the surrounding terrain, whereas a DEM is a regular 2.5-D raster that may be substantially coarser. At the resolved terrain scales, we model lunar topography as a single-valued elevation field $z = h(x, y)$ and use cubic B-splines to represent both inputs continuously. This common height-field representation permits evaluation away from the original sample locations and provides analytic terrain gradients for pose refinement.

A map-wide search must score millions of poses on a regular lattice, whereas metric accuracy requires continuous coordinates and the measured points themselves. We therefore split localization into two stages sharing one height-matching objective, invariant to an unknown constant vertical offset. In retrieval, its decomposition into masked cross-correlations enables FFT evaluation over the translation lattice at sampled headings. The retained candidates are then refined and re-ranked with the same objective on raw LiDAR points and the analytic gradient of the map height field. Roll and pitch are assumed available from an external attitude estimator.

Our contributions are:

- A common cubic B-spline height-field representation for a coarse orbital DEM and a single LiDAR scan, supporting evaluation and analytic gradients away from the source samples.
- An offset-invariant height objective that decomposes into masked cross-correlations, making a map-wide search over translations and headings tractable by FFT, and that is reused as the refinement cost on raw points.
- Evaluation against planetary and LiDAR localization baselines on four simulated lunar terrains and six DEM resolutions, with component ablations and a reference-map experiment using native orbital products.

II. RELATED WORK

Planetary terrain-map localization. Carle et al. [6] matched topographic peaks extracted from rover LiDAR against an orbital elevation map, with multi-frame odometric integration improving robustness. Geromichalos et al. [9] periodically registered an accumulated local elevation map to an orbital map within particle-filter SLAM, and Le Gentil et al. [17] used Gaussian-process gradient maps for loop closure in planetary-analog terrain. Zhang et al. [16] incorporated a DEM-derived fitted surface into LiDAR-inertial SLAM through Monte Carlo localization initialized within a supplied region of interest. These methods depend on repeatable terrain features, temporal accumulation, or a bounded initialization region. In contrast, SplineLoc targets map-wide localization from a single LiDAR scan without terrain-landmark extraction, temporal accumulation, or a bounded position initialization.

Rover-to-orbital visual and landmark methods. Visual methods relate surface imagery to orbital imagery through geometric alignment [10] or learned lunar cross-view matching [11], [13]; Censible [12] localizes Mars panoramas against orbital imagery given an externally estimated orientation and an odometry-bounded search region. ShadowNav [7] associates observed craters with an orbital crater map, and Tian et al. [8] match horizon profiles predicted from a DEM. These methods rely on visual appearance or on geometric primitives that must remain recoverable across the rover and orbital views.

LiDAR global localization. General LiDAR global localization commonly separates place retrieval from metric pose estimation. RING [14] uses a Radon sinogram, BEVPlace++ [15] learned rotation-equivariant BEV features,

and CrossLoc3D [18] bridges aerial and ground LiDAR. These methods retrieve against a database of reference scans; SplineLoc instead evaluates candidate poses directly against a supplied DEM.

Continuous map representations. B-splines have been used as continuous representations in SLAM, including feature curves [19] and online scan-to-map surfaces [20]. We instead use a B-spline height-field representation to place a pre-existing orbital DEM and a single rover LiDAR scan in a common continuous form for global localization.

III. PROPOSED METHOD

A. Overview

SplineLoc estimates the planar pose of a single LiDAR scan relative to a georeferenced DEM by comparing terrain heights (Fig. 2). We fit the DEM offline and the levelled scan online as cubic B-spline height fields (Sec. III-C). For *retrieval*, we sample these fields on a common lattice and define a height-matching objective that is invariant to an unknown constant vertical offset, hereafter the *offset-invariant objective*; expanding it into masked cross-correlations enables FFT evaluation over the translation lattice at each sampled heading (Sec. III-D). The retained candidates undergo continuous *refinement* and re-ranking using raw LiDAR points and the analytic map gradient (Sec. III-E). Both stages use the same offset-invariant least-squares form, but retrieval evaluates supported lattice cells, whereas refinement evaluates a range-stratified subset of measured points.

B. Problem Statement

The inputs are a DEM, i.e., elevations on a georeferenced grid of spacing r_D with a validity mask, and a single LiDAR scan levelled to the map vertical using externally supplied roll and pitch. We assume that roll and pitch are available from a calibrated external attitude estimator, such as an IMU aided by a star tracker for high-accuracy attitude determination [21], [22]. Only roll and pitch are used for scan levelling; position and heading are estimated through scan-map matching. Each LiDAR point is a pair (s_i, z_i) , where $s_i \in \mathbb{R}^2$ is its horizontal coordinate in the sensor frame and z_i is its vertical coordinate relative to the sensor origin. The unknown planar pose is $\xi = [c_x, c_y, \psi]^\top$, comprising east-north position $\mathbf{c} = [c_x, c_y]^\top$ and heading ψ . Under a candidate pose ξ , the horizontal map-frame position of point i is

$$\mathbf{p}_i(\xi) = \mathbf{R}_\psi \mathbf{s}_i + \mathbf{c}, \quad (1)$$

where $\mathbf{R}_\psi$ is the planar rotation matrix. Thus, $\mathbf{p}_i(\xi) \in \mathbb{R}^2$ denotes the location in the DEM at which the height corresponding to LiDAR point i is evaluated. The search domain is the full extent of the supplied DEM; no position or heading initialization is provided. We treat each scan as a rigid observation, assuming negligible or compensated motion distortion.

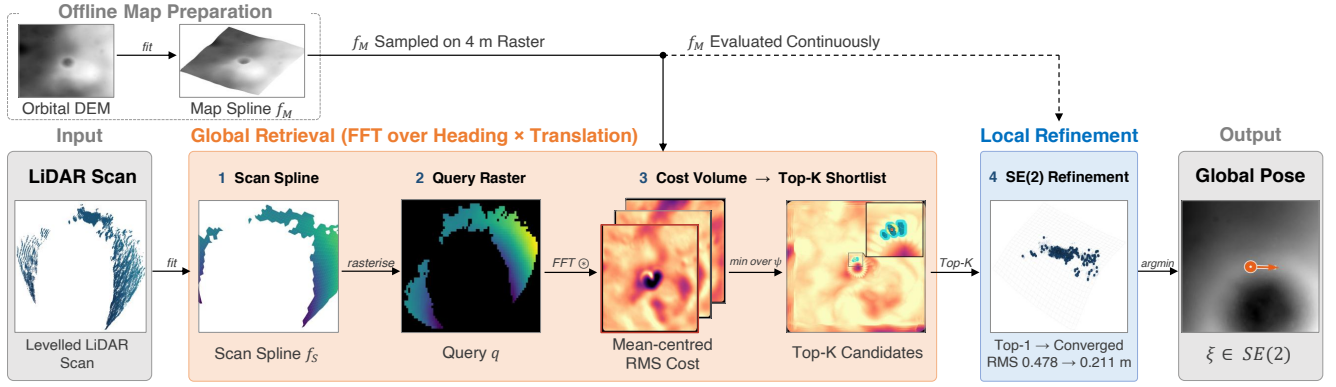


Fig. 2. **SplineLoc pipeline.** Offline, the DEM is fitted with a cubic B-spline height field f_M . Online, the levelled scan is fitted with f_S and sampled at N_ψ headings. The offset-invariant height objective is evaluated over the translation lattice by masked FFT; the K retained candidates are refined and re-ranked using raw LiDAR points and the analytic gradient of f_M .

C. Continuous Height-Field Representation

At the resolved terrain scale, we model both the DEM and the levelled LiDAR scan as single-valued height fields. Let β^3 denote the centred cubic B-spline kernel [23]. On a uniform knot grid of spacing h , the basis functions are $B_j(x) = \beta^3((x - x_j)/h)$, and the corresponding tensor-product height field is

$$f(x, y; \mathbf{C}) = \sum_{j,k} C_{jk} B_j(x) B_k(y), \quad (2)$$

where $\mathbf{C}$ is the coefficient grid. The compact support of the cubic basis limits each evaluation to 4×4 coefficients, while its C^2 continuity provides an analytic spatial gradient ∇f .

We use this representation for the map field f_M and scan field f_S . For the scattered LiDAR samples, the scan coefficients are obtained by penalized least squares [24],

$$\mathbf{C}_S = \arg \min_{\mathbf{C}} \sum_i [z_i - f(s_{ix}, s_{iy}; \mathbf{C})]^2 + \alpha_S \mathcal{R}(\mathbf{C}), \quad (3)$$

where $\mathcal{R}(\mathbf{C})$ is a second-difference smoothness penalty along both axes, suppressing poorly constrained local variation in sparsely sampled regions. For the regular-grid DEM, $\mathbf{C}_M$ is obtained by an equivalent separable penalized fit that exploits its grid structure [25].

The map knot spacing is $h_M = \max(h_0, r_D)$, avoiding a coefficient grid finer than the source DEM when $r_D > h_0$; otherwise the map is represented at the fixed scale h_0 . The scan spacing h_S is fixed. For retrieval, f_M and f_S are sampled on a common lattice, whereas refinement evaluates f_M and ∇f_M directly at the transformed LiDAR points.

D. Stage 1: Global Retrieval by Masked Height Correlation

We first discretize the continuous map and scan fields on a common two-dimensional lattice of spacing Δ . Let $\mathbf{u} = [u_x, u_y]^\top \in \mathbb{Z}^2$ denote an integer lattice index, and let $\mathbf{o}_M \in \mathbb{R}^2$ denote the metric position of the map-lattice origin. Sampling the map spline gives

$$F(\mathbf{u}) = f_M(\mathbf{o}_M + \Delta \mathbf{u}),$$

where F is the map-height raster. The binary mask $V(\mathbf{u}) \in \{0, 1\}$ indicates whether the corresponding map location is

valid in the source DEM; we set $F(\mathbf{u}) = 0$ wherever $V(\mathbf{u}) = 0$.

For each sampled heading ψ , we construct a sensor-centred query raster on a lattice with the same spacing Δ . A query-lattice index $\mathbf{u}$ corresponds to the horizontal scan-frame location $\Delta \mathbf{R}_\psi^\top \mathbf{u}$. We define the binary support mask $m_\psi(\mathbf{u})$ to be one when this location lies within horizontal distance ρ of at least one measured LiDAR point $\mathbf{s}_i$, and zero otherwise. The query raster is then

$$q_\psi(\mathbf{u}) = m_\psi(\mathbf{u}) f_S(\Delta \mathbf{R}_\psi^\top \mathbf{u}). \quad (4)$$

Thus, m_ψ specifies which query cells are supported by the measured scan, while q_ψ stores the corresponding spline heights.

For a fixed heading ψ , let $\boldsymbol{\tau} \in \mathbb{Z}^2$ denote a candidate translation of the query raster over the map lattice. Under this translation, query cell $\mathbf{u}$ is compared with map cell $\mathbf{u} + \boldsymbol{\tau}$. A cell contributes to the matching cost only when it belongs to the query support and overlaps a valid map cell. The number of such overlapping cells is therefore

$$n_\psi(\boldsymbol{\tau}) = \sum_{\mathbf{u}} m_\psi(\mathbf{u}) V(\mathbf{u} + \boldsymbol{\tau}). \quad (5)$$

The DEM and the levelled sensor frame may differ by an unknown constant vertical offset b . We therefore score each candidate $(\boldsymbol{\tau}, \psi)$ by the minimum mean-squared height discrepancy after eliminating this offset:

$$E_{\text{ret}}^2(\boldsymbol{\tau}, \psi) = \min_b \frac{1}{n_\psi(\boldsymbol{\tau})} \sum_{\mathbf{u}} m_\psi(\mathbf{u}) V(\mathbf{u} + \boldsymbol{\tau}) \cdot [q_\psi(\mathbf{u}) - F(\mathbf{u} + \boldsymbol{\tau}) - b]^2. \quad (6)$$

The minimizing b is the mean height residual over the valid overlap. Hence, the objective compares relative terrain-height variation rather than requiring the absolute vertical levels of the scan and DEM to coincide.

Directly evaluating Eq. (6) independently for every translation would repeatedly sum over the query support. Instead, after substituting the optimal b and expanding the squared residual, all required sums take the form of cross-correlations. We define

$$(A \star B)(\boldsymbol{\tau}) = \sum_{\mathbf{u}} A(\mathbf{u}) B(\mathbf{u} + \boldsymbol{\tau}),$$

where $\star$ denotes discrete cross-correlation. In particular, $n_\psi = m_\psi \star V$, and the retrieval objective becomes

$$E_{\text{ret}}^2(\tau, \psi) = \frac{(q_\psi^2 \star V) - 2(q_\psi \star F) + (m_\psi \star F^2)}{n_\psi} - \left[\frac{(q_\psi \star V) - (m_\psi \star F)}{n_\psi} \right]^2, \quad (7)$$

where every correlation and n_ψ in Eq. (7) is evaluated at τ .

Equation (7) therefore reduces evaluation over all translations to a small set of cross-correlations. By the correlation theorem,

$$A \star B = \mathcal{F}^{-1} \left[\overline{\mathcal{F}[A]} \mathcal{F}[B] \right],$$

so these correlations are evaluated efficiently by FFT for each sampled heading. Zero-padding is used to remove circular wrap-around [26], and the map-side spectra of V , F , and F^2 are cached across headings.

To prevent candidates supported by only a small portion of the query from receiving spuriously low scores, translations satisfying

$$n_\psi(\tau) < \eta \sum_{\mathbf{u}} m_\psi(\mathbf{u})$$

are discarded. Among all remaining translation-heading pairs, the K candidates with the lowest E_{ret} are retained for refinement. Finally, a retained lattice translation τ is converted to the metric map position

$$\mathbf{c} = \mathbf{o}_M + \Delta\tau.$$

E. Stage 2: Continuous Refinement on Raw LiDAR Points

Retrieval quantizes the initial position and heading and scores a smoothed scan field. We refine each candidate in continuous pose coordinates using measured LiDAR points rather than the scan spline. To limit the influence of the densely sampled near range, we draw a fixed subset of N_{ref} points using equal per-band quotas from concentric horizontal-range bands; the same subset is used for every candidate and iteration of a scan.

For point i at pose ξ , define the height residual and its mean as

$$r_i(\xi) = z_i - f_M(\mathbf{p}_i(\xi)), \quad \bar{r}(\xi) = \frac{1}{N_{\text{ref}}} \sum_i r_i. \quad (8)$$

Eliminating the best constant offset gives

$$E_{\text{ref}}^2(\xi) = \frac{1}{N_{\text{ref}}} \sum_{i=1}^{N_{\text{ref}}} (r_i - \bar{r})^2. \quad (9)$$

This is the offset-invariant form of Eq. (6) evaluated on measured points, but its point-based sampling differs from retrieval's supported lattice cells. The minimizers therefore need not coincide, and candidates are re-ranked after refinement.

For evaluations in the map interior, differentiating through $\mathbf{p}_i(\xi)$ gives the 1×3 Jacobian

$$\mathbf{J}_i = -\nabla f_M(\mathbf{p}_i)^\top [\mathbf{I}_2 \quad \dot{\mathbf{R}}_\psi \mathbf{s}_i], \quad (10)$$

where $\dot{\mathbf{R}}_\psi = \partial \mathbf{R}_\psi / \partial \psi$. The centred residual $\tilde{r}_i = r_i - \bar{r}$ has Jacobian $\tilde{\mathbf{J}}_i = \mathbf{J}_i - \frac{1}{N_{\text{ref}}} \sum_j \mathbf{J}_j$. Stacking these into $\tilde{\mathbf{r}}$ and $\tilde{\mathbf{J}}$, each Gauss-Newton increment solves $\delta \xi \in \arg \min_\delta \|\tilde{\mathbf{r}} + \tilde{\mathbf{J}}\delta\|_2^2$, followed by fixed component-wise step limits before the pose, residuals, and Jacobians are updated. After refinement terminates for all candidates, the pose with the lowest final E_{ref} is returned; it is not restricted to the retrieval lattice.

F. Implementation Details

Unless otherwise stated, we use $h_0 = h_S = \Delta = 4$ m, $\rho = 3$ m, $N_\psi = 24$, $\eta = 0.85$, and $K = 100$. For spline fitting, the nominal regularization parameters are $\lambda_M = \lambda_S = 10^{-3}$. Their effective penalty weights are scaled to the corresponding data and penalty matrices as

$$\alpha = \lambda \frac{\text{tr}(\mathbf{H})}{\max(\text{tr}(\mathbf{P}), 10^{-12})},$$

where $\mathbf{H}$ is the data Gram matrix and $\mathbf{P}$ is the second-difference penalty matrix. This scaling is applied separately to each DEM axis and to the two-dimensional scan fit. Missing DEM cells are median-filled for fitting but remain invalid in V .

Refinement samples up to 80 points per horizontal-range band with boundaries $\{0, 5, 15, 30, 60, 250\}$ m and a fixed seed, giving $N_{\text{ref}} \leq 400$, and runs at most $I = 60$ iterations per candidate with component-wise step limits of 10 m in translation and 0.25 rad in heading, stopping when both translation increments have magnitude below 10^{-4} m.

IV. EVALUATION

A. Lunar Dataset and Evaluation Protocol

We used a lunar simulation dataset constructed with OmniLRS [28]. The four environments were initialized from real orbital topography: Apollo 11, Apollo 14, and Apollo 15 from 2 m/pixel LROC NAC DTMs [4], and Haworth from a 1 m/pixel shape-from-shading (SfS) DEM [5]. The supplied reference maps span 300×320 m at Apollo 11, 2000×2000 m at Apollo 14 and Apollo 15, and 400×400 m at Haworth. The sites cover contrasting terrain: low-relief mare at Apollo 11, crater relief at Apollo 14, directional relief near Hadley Rille at Apollo 15, and extended crater-wall slopes at Haworth.

The simulated rover carried a 128-channel LiDAR with a 360° horizontal field of view. Scans were rendered instantaneously, and the scan sequence was subsampled at approximately 1 m intervals between acquisition positions. We selected 100 queries per site and kept them fixed across all methods and resolutions. No odometry or temporal accumulation was used. SplineLoc levelled each scan using simulator-provided roll and pitch and received no position or heading initialization.

Each simulated terrain was exported at 0.25 m/pixel and area-mean downsampled to 0.5, 1, 2, 5, and 10 m/pixel reference maps. Thus, 0.25 m/pixel denotes the simulation-terrain sampling rather than an orbital observation; Sec. IV-E separately evaluates the native orbital products. We report median horizontal error $e_{xy} = \|\hat{\mathbf{c}} - \mathbf{c}\|_2$, success rate within

TABLE I

LOCALIZATION PERFORMANCE ACROSS DEM RESOLUTIONS. MEDIAN IS THE HORIZONTAL POSITION ERROR AND SR@5M THE SUCCESS RATE WITHIN 5 M. #EST. IS THE NUMBER OF ESTIMATES RETURNED BY DARCES; ITS MEDIAN IS COMPUTED OVER RETURNED ESTIMATES ONLY AND EXCLUDED FROM RANKING, WHILE MISSING ESTIMATES COUNT AS FAILURES IN SR@5M. BEST VALUE PER METRIC AND ROW IS SHADED.

Site	DEM res. (m/pixel)	DARCES [6]			BEVPlace++ [15]		RING [14]		ICP [27]		SplineLoc (Ours)	
		Median (m)	SR@5m (%)	#Est.	Median (m)	SR@5m (%)	Median (m)	SR@5m (%)	Median (m)	SR@5m (%)	Median (m)	SR@5m (%)
Apollo 11	0.25	—	0.0	0/100	101.25	15.0	58.06	24.0	84.97	1.0	1.55	98.0
	0.5	5.63	1.0	3/100	101.86	15.0	50.81	29.0	83.30	1.0	1.55	98.0
	1	3.53	12.0	18/100	101.99	13.0	51.68	28.0	85.49	0.0	1.55	98.0
	2	5.49	12.0	26/100	102.14	10.0	59.71	26.0	85.59	0.0	1.54	98.0
	5	5.84	15.0	30/100	104.55	10.0	66.86	23.0	82.95	1.0	1.62	98.0
	10	40.54	0.0	9/100	102.98	1.0	65.75	5.0	95.88	1.0	1.86	96.0
Apollo 14	0.25	2.64	2.0	2/100	1.49	78.0	1.66	74.0	52.02	0.0	1.05	98.0
	0.5	2.35	37.0	38/100	1.40	84.0	1.48	77.0	52.03	0.0	1.05	98.0
	1	2.86	40.0	44/100	1.61	85.0	1.68	80.0	52.27	0.0	1.05	98.0
	2	3.54	22.0	31/100	2.33	85.0	2.42	79.0	52.24	1.0	1.05	98.0
	5	5.36	9.0	24/100	41.44	28.0	8.40	25.0	63.79	0.0	1.03	99.0
	10	—	0.0	0/100	220.79	0.0	171.82	1.0	91.44	2.0	1.22	96.0
Apollo 15	0.25	1.39	35.0	43/100	0.89	82.0	1.23	88.0	570.67	0.0	1.10	100.0
	0.5	1.52	71.0	84/100	0.90	83.0	1.23	90.0	557.28	0.0	1.10	100.0
	1	1.54	88.0	97/100	1.12	84.0	1.46	87.0	574.82	0.0	1.10	100.0
	2	1.99	80.0	94/100	2.06	71.0	1.78	91.0	612.56	0.0	1.10	100.0
	5	4.69	46.0	83/100	418.10	19.0	4.55	55.0	557.44	0.0	1.18	100.0
	10	318.65	0.0	58/100	548.23	0.0	458.35	2.0	535.28	0.0	1.84	97.0
Haworth	0.25	—	0.0	0/100	44.64	8.0	16.69	31.0	92.62	0.0	1.56	83.0
	0.5	12.88	4.0	19/100	59.24	7.0	22.18	28.0	92.64	0.0	1.56	82.0
	1	52.38	4.0	44/100	53.99	5.0	20.47	27.0	92.66	0.0	1.56	82.0
	2	145.35	1.0	41/100	58.84	2.0	28.10	22.0	90.43	0.0	1.56	81.0
	5	227.54	0.0	6/100	60.79	2.0	48.60	14.0	92.43	0.0	1.58	81.0
	10	—	0.0	0/100	63.98	3.0	50.82	3.0	87.41	0.0	1.79	78.0

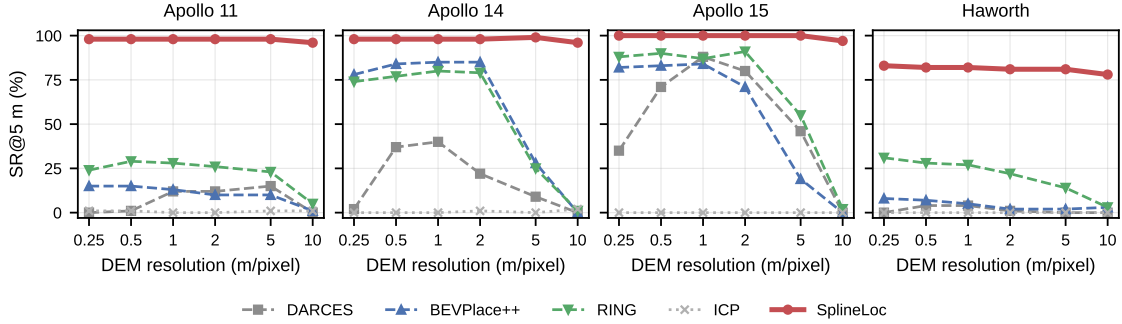


Fig. 3. **Localization success rate (SR@5m) across DEM resolutions at four lunar sites.** Each method is evaluated on the same 100 query scans per site at all six resolutions. The 5 m success threshold is held fixed throughout the sweep.

5 m (SR@5m), and, for SplineLoc, median heading error e_ψ . Missing estimates count as failures. The same SplineLoc configuration was used at every resolution, with only $h_M = \max(4 \text{ m}, r_D)$ changing with the DEM.

B. Baselines

We compared SplineLoc with DARCES [6], BEVPlace++ [15], RING [14], and multi-start ICP [27]. DARCES was reimplemented from the published formulation because no official code was available to us; the other baselines used official code. For each baseline, we tested unlevelled scans and scans levelled with simulator-provided roll and pitch, reporting the better-performing variant. Among DARCES, DARCES followed by RANSAC, and the complete single-frame MOGA pipeline, DARCES performed best and is reported.

BEVPlace++ was fine-tuned on Apollo 14 LiDAR-DEM pairs with training runs separate from the held-out query run; the same checkpoint was used at all sites. RING retained its sinogram representation and fixed $z > -0.6 \text{ m}$ filter on query and reference clouds. As a local-registration control,

ICP used ten uniformly sampled planar initializations with zero rotation and returned the highest-fitness result.

BEVPlace++ and RING were evaluated after their complete pipelines, including pose refinement. Their reference views were rendered from the DEM at 5 m intervals using a virtual LiDAR (height 0.415 m, 128×2048 rays, first intersections), not a prior traverse. BEVPlace++ encoded 201×201 count-BEV images at 0.4 m/pixel with REIN into 8192-dimensional descriptors. At Apollo 14/15, the databases covered the query-trajectory bounding box expanded by 40 m (BEVPlace++) or 80 m (RING); at Apollo 11 and Haworth, they covered the supplied DEM. SplineLoc searched the full supplied DEM subject to its overlap criterion, a larger domain at Apollo 14/15.

C. Localization Performance Across Resolutions and Terrains

Table I and Fig. 3 show that SplineLoc achieved the highest SR@5m in all 24 site-resolution conditions, with median horizontal errors of 1.03–1.86 m. Across the four sites, mean SR@5m decreased from 94.75% at 0.25 m/pixel

TABLE II

COMPONENT ABLATIONS POOLED OVER THE FOUR SITES (400 QUERIES PER CONDITION). MED. AND e_ψ ARE THE FINAL MEDIAN HORIZONTAL (M) AND HEADING (DEG) ERRORS. $R@K$ IS PRE-REFINEMENT RETRIEVAL RECALL: AT LEAST ONE OF THE K CANDIDATES IS WITHIN 5 M AND 15° OF GROUND TRUTH. IT MEASURES SHORTLIST COVERAGE AND IS NOT DIRECTLY COMPARABLE TO FINAL SR@5M.

Condition	2 m/pixel					5 m/pixel					10 m/pixel			
	Med.	SR@5m	e_ψ	$R@K$		Med.	SR@5m	e_ψ	$R@K$		Med.	SR@5m	e_ψ	$R@K$
Default	1.26	94.2	0.33	96.2		1.31	94.5	0.36	95.0		1.71	91.8	0.57	94.0
No refinement	2.85	78.5	0.87	96.2		2.92	77.0	0.87	95.0		3.04	75.0	0.87	94.0
$N_\psi = 12$	1.33	88.0	0.38	75.5		1.42	88.2	0.42	73.5		1.81	85.5	0.67	71.8
$N_\psi = 48$	1.26	94.2	0.33	98.2		1.31	94.5	0.36	97.2		1.71	91.8	0.57	95.8
$h_M = 4$ m	1.26	94.2	0.33	96.2		1.32	94.0	0.36	95.0		1.72	91.0	0.54	94.0
$\eta = 0.60$	1.26	93.5	0.33	96.2		1.31	93.8	0.36	95.8		1.71	91.0	0.57	96.0

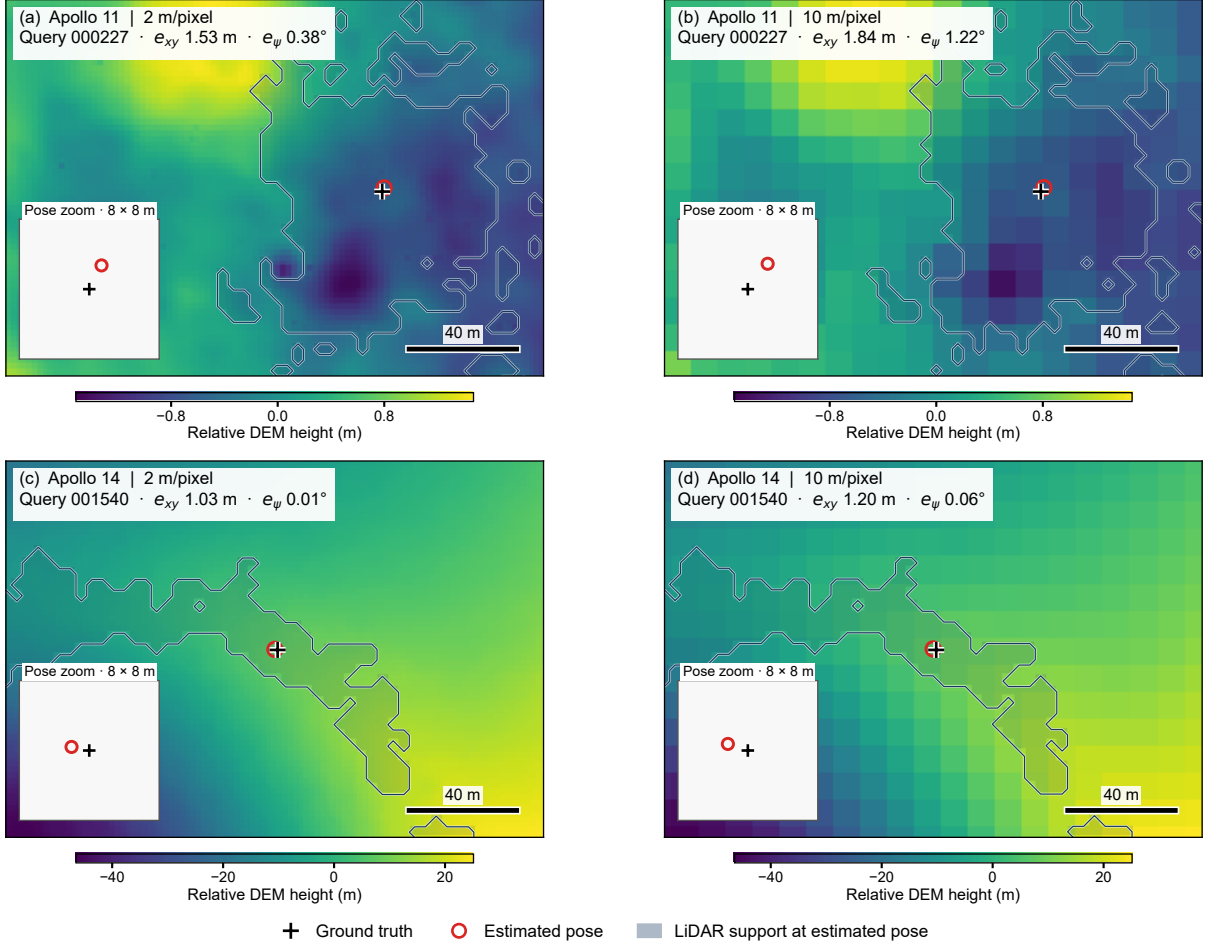


Fig. 4. **Qualitative localization across DEM resolutions.** Within each row, the same query is localized independently against simulation-derived reference maps at 2 and 10 m/pixel, with a fixed crop and height colour scale. Backgrounds show source DEM cells without display smoothing. Blue shading and outlines indicate LiDAR support at the estimated pose; crosses and red circles mark ground-truth and estimated positions. Insets span 8×8 m. Crops are for visualization only; localization searches the full supplied DEM.

to 91.75% at 10 m/pixel. The map and scan splines and retrieval lattice retain a 4 m nominal scale for 0.25–2 m/pixel maps. At 5 and 10 m/pixel, the map knot spacing follows the DEM resolution, while matching uses relative heights over the scan footprint.

This behavior contrasts with the database baselines. At 5 and 10 m/pixel, SplineLoc achieved 94.50% and 91.75% SR@5m, whereas RING, the strongest baseline at these resolutions, achieved 29.25% and 2.75%. BEVPlace++ and RING remained competitive on Apollo 14 and Apollo 15

at finer resolutions, but their performance degraded sharply as the reference map was coarsened. The results therefore suggest that SplineLoc relies less on fine-scale local geometry and instead matches broader height variation preserved across DEM resolutions.

DARCES illustrates the distinction between estimate availability and accuracy (Table I). At Apollo 14 and 0.25 m/pixel, its 2.64 m median was computed from only two returned estimates out of 100 queries, with 2% SR@5m. At Apollo 15, it achieved 88% SR@5m at 1 m/pixel, but returned 58

TABLE III

REFERENCE MAPS FROM NATIVE ORBITAL PRODUCTS. QUERY SCANS AND CONFIGURATION ARE THOSE OF TABLE I; ONLY THE MAP IS REPLACED BY THE NATIVE LROC NAC DTM (APOLLO, 2 M/PIXEL) OR SfS DEM (HAWORTH, 1 M/PIXEL) AND ITS AREA-MEAN DOWNSAMPLING TO 5 AND 10 M/PIXEL, AFTER A FIXED METADATA-BASED SITE-WISE TRANSLATION SHARED BY ALL QUERIES.

Site	r_D	Sim.-derived		Native product		e_ψ
		Med.	SR@5m	Med.	SR@5m	
Apollo 11	2	1.54	98.0	1.54	98.0	0.70
	5	1.62	98.0	1.64	98.0	0.70
	10	1.86	96.0	1.87	95.0	1.05
Apollo 14	2	1.05	98.0	1.05	98.0	0.19
	5	1.03	99.0	1.04	98.0	0.23
	10	1.22	96.0	1.21	97.0	0.31
Apollo 15	2	1.10	100.0	1.08	100.0	0.23
	5	1.18	100.0	1.22	100.0	0.28
	10	1.84	97.0	1.80	98.0	0.59
Haworth	1	1.56	82.0	1.57	82.0	0.59
	5	1.58	81.0	1.59	79.0	0.69
	10	1.79	78.0	1.80	78.0	0.77

estimates with no successes at 10 m/pixel. Thus, failures include both missing estimates and returned poses outside the success threshold. The median over returned estimates should therefore be interpreted alongside SR@5m and #Est., rather than as a standalone measure of localization reliability.

Apollo 14 and Apollo 15 provide the main evidence for map-wide retrieval because both reference maps span 2000×2000 m. At 10 m/pixel, SplineLoc still achieved 96% and 97% SR@5m with median errors of 1.22 and 1.84 m, respectively. The final estimates are not restricted to DEM cells or the retrieval lattice because refinement evaluates the continuous map field directly. Thus, sub-cell position accuracy reflects continuous alignment of terrain structure over the scan footprint rather than recovery of terrain detail absent from the coarse DEM.

Apollo 11 represents a different regime: its 300×320 m map is comparable to the scan footprint and contains relatively weak low-relief structure. Nevertheless, SplineLoc maintained 96–98% SR@5m with median errors below 2 m, indicating that a prominent extracted landmark is not required when sufficient distributed height variation is available.

Haworth remained the most challenging site, with SR@5m decreasing from 83% to 78% across the resolution sweep. Its extended crater-wall segments contain similar local surface normals. On a locally planar region, translation primarily introduces an approximately constant height change, which is removed by the offset-invariant objective. Large slope magnitude alone is therefore not necessarily distinctive; spatial variation in terrain gradient is more informative for resolving position.

Qualitative localization. Figure 4 shows two queries with horizontal errors near the corresponding site-wise medians in Table I. At 10 m/pixel, broad height patterns remain visible,

with horizontal errors of 1.84 and 1.20 m and heading errors of 1.22° and 0.06° .

D. Component Ablations

Table II evaluates refinement, heading sampling, map scale, and overlap. R@K measures pre-refinement shortlist coverage; SR@5m evaluates the final estimate.

Without refinement, SR@5m drops from 91.8–94.5% to 75.0–78.5%, with larger position and heading errors, supporting continuous refinement and re-ranking of retrieval candidates.

With $N_\psi = 12$, lower R@K and SR@5m indicate missed hypotheses. Using $N_\psi = 48$ slightly improves R@K but not final SR@5m, so we retain $N_\psi = 24$ to avoid approximately doubling the heading-wise FFT workload without improving localization.

Fixing $h_M = 4$ m changes SR@5m by at most 0.8 percentage points; we retain $h_M = \max(4 \text{ m}, r_D)$ to avoid a coefficient grid finer than a coarse DEM. Lowering η to 0.60 slightly increases R@K without improving SR@5m, supporting $\eta = 0.85$ to reject weak-overlap matches.

E. Reference Maps from Native Orbital Products

The resolution sweep used maps exported from the query-generating terrain, including procedural craters. We replaced them with native orbital products (Apollo: 2 m/pixel LROC NAC DTMs; Haworth: 1 m/pixel SfS DEM) and their area-mean downsampling to 5 and 10 m/pixel, keeping queries and configuration fixed. Each native map was cropped and transformed to the simulation frame by a site-wise horizontal translation derived solely from coordinate and crop metadata, not query scans or their ground-truth poses. This translation was shared across all queries and resolutions.

SR@5m changed by at most two percentage points at every site and resolution (Table III). Both map variants share the underlying orbital terrain; only the simulation-derived maps contain the added craters. The similar performance suggests that the shared terrain structure remains informative despite these differences.

F. Computational Cost

Timings were averaged over eight queries per site in an isolated single-process CPU run (Intel Core i5-13400F, 64 GB DDR5-5600, no GPU). Offline map fitting takes at most 0.22 s (2.5×10^5 coefficients at Apollo 14/15, 2 m/pixel). Online runtimes are 1.6 s at Apollo 11, 4.5–5.1 s at Apollo 14/15, and 5.7–5.9 s at Haworth, with little variation across DEM resolutions. Scan fitting dominates (0.7–4.9 s, increasing with point count); 24-heading FFT retrieval takes 0.15–1.24 s depending on map area, and refinement of $K = 100$ candidates takes 0.5–1.3 s. Database baselines also require pseudo-LiDAR reference rendering: about 2 h per resolution for trajectory-restricted Apollo 15. SplineLoc targets periodic global fixes rather than per-frame estimation, allowing stationary scan acquisition.

V. CONCLUSION

We presented SplineLoc, a map-wide single-scan LiDAR-to-DEM localization method based on a common cubic B-spline height-field representation. An offset-invariant height objective enables masked FFT retrieval over lattice translations and sampled headings, followed by continuous refinement and re-ranking on raw LiDAR points. The method requires no position or heading initialization, terrain-landmark extraction, or reference-scan database.

On four simulated lunar terrains and six DEM resolutions from 0.25 to 10 m/pixel, SplineLoc achieved the highest SR@5m in all 24 site-resolution conditions. At 10 m/pixel, mean SR@5m remained 91.75%, compared with 2.75% for the strongest baseline at that resolution. The method also maintained high localization performance over the 2000 × 2000 m Apollo 14 and Apollo 15 maps, demonstrating robustness both to DEM coarsening and to map-wide search over large terrain regions. Replacing the simulation-derived reference maps with aligned native orbital products changed SR@5m by at most two percentage points, further indicating that the method relies on terrain structure preserved across map representations rather than fine-scale details specific to the simulated maps. Future work will focus on resolving ambiguous terrain such as extended crater-wall regions with similar local surface normals, where weak spatial variation in terrain gradients can make candidate discrimination and re-ranking difficult.

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